• THE PROBLEM

Agriculture Still Runs on Guesswork

- 💧 Water is often wasted through reactive irrigation
- 🌱 Plant stress is detected too late
- 📶 Smart systems fail without stable internet
- 📈 Food demand continues to rise

Traditional farming needs intelligent automation

• THE SOLUTION

Integrated Offline Edge AI Farming System

Soil Sensor

+

Camera

+

BME280

→

Raspberry Pi 4

Local AI Brain

Runs entirely on-device

→

Predictive Irrigation

→ Pump Control

Vision AI

→ Dashboard

✓ Predicts watering needs

✓ Detects plant stress

✓ Works without internet — intelligence is processed locally, not remotely

HOW IT WORKS

Two AI Engines, One Device

Predictive Irrigation

- Analyzes soil moisture, temperature & humidity from BME280 sensor
- On-device ML model predicts when plants will need water
- Schedules irrigation proactively — not reactively

Pump Control

Vision AI

- Captures crop images via on-site camera at regular intervals
- On-device computer vision detects early signs of plant stress & disease
- Alerts farmer instantly via dashboard — no cloud delay

Dashboard

Both engines run simultaneously — on a single Raspberry Pi 4

Real Impact for Nigeria and Beyond

Saves Water

Precision delivery, zero waste

Improves Yield

Healthier crops, higher output

No Internet Required

Edge AI works offline anywhere

Low Cost

Affordable hardware for all

Scalable

From one plot to entire farms

Smart agriculture should be accessible, not exclusive.

• WHY THIS WINS

Built for Impact.
Built for the Future.

- AI
- Embedded Systems
- Automation
- Sustainability
- Agriculture
- Scalability

Engineering intelligence where it is needed most.

Thank You

Engineering intelligence where it is needed most.